Red Eye

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Introduction

A red eye is common in a wide variety of ocular conditions and signals inflammation. The majority of these conditions are benign, and many are self-limited, but some can have serious consequences. For example, conjunctivitis can be the initial presenting sign of COVID-19, which can be potentially fatal.[1] Table 1 lists minor and major/serious causes of red eye. The presence of 1 or more warning signs (see Table 2) requires immediate referral to an ophthalmologist or an optometrist if an ophthalmologist is not available.

Conjunctivitis is the most common cause of red eye.[2] Differentiating between the various causes of acute conjunctivitis is clinically very difficult with few tests to help. The majority of infectious (caused by viruses or bacteria) or noninfectious (due to an allergic reaction or drug toxicity) cases of conjunctivitis can be treated in primary care but certain cases require referral to a specialist (see Table 3).

Table 1: *Etiology of Red Eye*

Cause	Examples
Infection	• Conjunctivitis (pink eye) ◦ Acute (e.g., bacterial, viral) ◦ Chronic (nasolacrimal duct obstruction) • Keratitis (bacterial, viral, fungal, parasitic, other [contact lens–related]) • Blepharitis with secondary conjunctivitis/keratitis
Allergy	Seasonal, keratoconjunctivitis (atopic, vernal), contact lens–related (giant papillary conjunctivitis or GPC)
Ocular inflammation	Iritis, uveitis, episcleritis, scleritis, corneal infiltrate/ulcer Related to systemic diseases (e.g., Crohn disease, ulcerative colitis, rheumatoid arthritis)
Glaucoma	Acute angle-closure glaucoma (see Glaucoma)
Dry eye disease (keratoconjunctivitis sicca)	Sjögren syndrome, vitamin A deficiency, meibomian gland dysfunction (MGD)/ocular rosacea (see Rosacea), other
Eyelid condition	Blepharitis, hordeolum (stye), chalazion, meibomian gland dysfunction; entropion, ectropion, lagophthalmos
Toxic/chemical/other irritants	Ophthalmic medications, contact lens solutions, acids/alkalis, smoke, wind, UV light (e.g., tanning bed, welder's arc)
Traumatic injury	Corneal abrasions, foreign bodies, blunt trauma with hyphema (bleeding into the anterior chamber), heat exposure
Ocular manifestations of skin condition	• Herpes zoster, herpes simplex (with lid, conjunctival and/or corneal involvement) • Rosacea (blepharitis, dry eye; see Rosacea) • Steven Johnson syndrome, psoriasis
Other	Subconjunctival hemorrhage, inflamed pterygium or pinguecula

Goals of Therapy

- Preserve eyesight
- Prevent structural damage to the eye
- Control and prevent spread of infection
- Control inflammation
- Provide symptomatic relief

Investigations

An algorithm for assessment and management of red eye is presented in Figure 1 and Figure 2. The presence of 1 or more warning signs (see Table 2) requires immediate referral to an ophthalmologist.

Thorough history to determine:

- Unilateral or bilateral eye involvement
- Duration and type of symptoms
- Type of discharge
- Visual changes
- Pain and photophobia

- Allergies
- Contact lens wear
- Trauma
- Ocular and systemic medications; history of drug use, smoking/vaping
- Previous eye diseases and treatment
- Other relevant medical history, including recent upper respiratory tract infection, fever, chills or rash; dermatologic and autoimmune conditions

Physical examination:

- Best corrected visual acuity with pinhole if glasses are not available
- Pupil size and reaction to light
- Inspection of face, eyelids, conjunctiva (to determine type and extent of hyperemia), sclera, cornea (staining with fluorescein)
- Globe tenderness by digital pressure through the lids
- Preauricular adenopathy
- Intraocular pressure if acute glaucoma is suspected

Table 2: *Warning Signs for Ophthalmologist Referral*

- **Marked redness of the eye**—limbal/ciliary injection (redness dominant at the corneoscleral junction), especially unilateral involvement
- **White corneal opacity** or corneal haze (with or without fluorescein staining)
- **Moderate/severe pain with photophobia or globe tenderness**—not relieved by test dose of topical anesthetic drop (proparacaine, tetracaine); possible acute angle-closure glaucoma (more common in older adults), corneal erosion/infiltrate/ulcer, iritis or scleritis
- **Reduced visual acuity**—Snellen chart using a pin hole; possible iritis, corneal ulcer, glaucoma (coloured halos around lights)
- **Pupil abnormalities** (miotic or mid-dilated and fixed, irregular, sluggish to light, painful pupillary constriction)
- **Recent trauma to eye** including ocular surgery, foreign body (e.g., hammering metal on metal), chemical exposure
- **Contact lens wear** (corneal infiltrate/ulcer)
- **History of iritis/scleritis/angle-closure glaucoma**—check for underlying systemic disease; history of contact lens–related ulcer, recurrent corneal erosion, chronic conjunctivitis (unilateral or bilateral) with duration >3 weeks
- **Suspected serious infectious causes:**
  - COVID-19—conjunctivitis alone or with fever, cough and shortness of breath
  - Chlamydia—preauricular node and purulent conjunctivitis plus history of sexual contact
  - Gonococcal—hyperacute conjunctivitis
  - Herpes simplex—usually unilateral, watery discharge and preauricular node (differential diagnosis is viral, i.e., adenovirus), eyelid vesicles
  - Herpes zoster—ocular involvement (usually eyelids, conjunctiva), facial vesicles, corneal complication, uveitis
  - Neonatal conjunctivitis—usually due to Chlamydial or Gonococcal infection; may also occur due to chemical toxicity secondary to prophylactic medications instilled at birth

Table 3: *Diagnosing Conjunctivitis*

History	Exposure to person with red eye Upper respiratory tract infection (URTI) Exposure to a person with possible COVID-19 Past history of conjunctivitis, blepharitis, styes Discharge, type and duration, purulent with morning crusting or watery History of atopic allergies, atopic disease Exposure to ophthalmic medications, chemicals Contact lens wear
Signs	Normal pupillary reaction Discharge and/or excessive tearing Lid and conjunctival edema Conjunctival redness Distinguishing bacterial from adenoviral conjunctivitis is clinically very difficult but a point-of-care adenoviral test, AdenoPlus, is available for health-care practitioners to use in office as a helpful diagnostic test;[3] Viral: more common in adults; often presents as a very red eye with swollen lids, conjunctival membranes and subconjunctival hemorrhages Bacterial: more common in children and is usually associated with a purulent discharge, often from one eye;[4][5] common pathogens seen in adults and children include *Haemophilus influenzae*, *Staphylococcus aureus* and *Streptococcus pneumoniae*; *Moraxella catarrhalis* may occasionally cause conjunctivitis, mainly in the pediatric population[5]
Refer to ophthalmologist	See Table 2

Photo 1: *Herpetic Blepharoconjunctivitis*

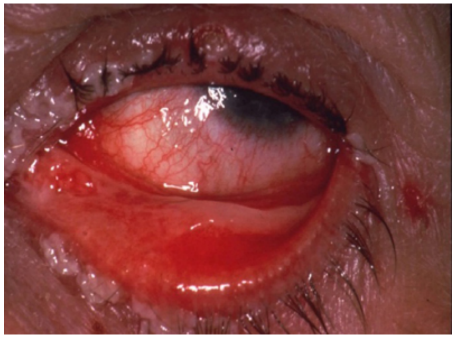

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Photo 2: *Acute Glaucoma (fixed pupil)*

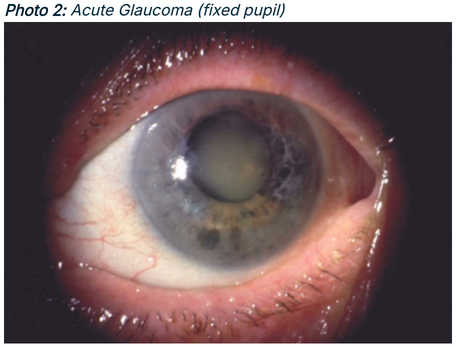

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Photo 3: *Contact Lens Infiltrate*

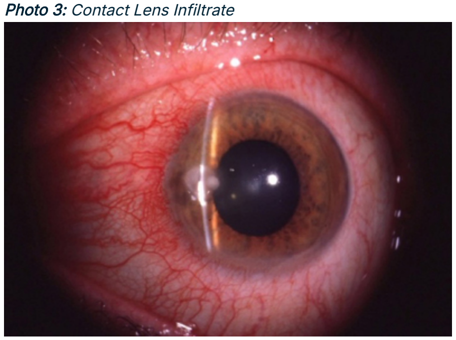

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Photo 4: *Pseudomonas Ulcer*

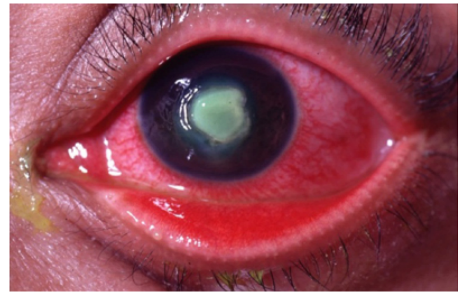

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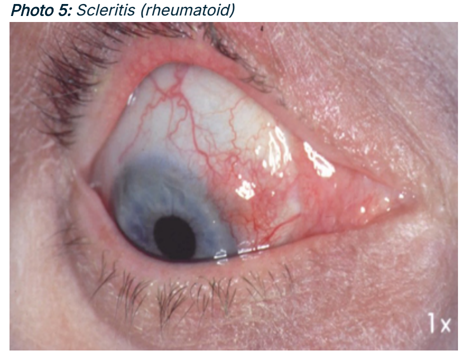
Photo 5: *Scleritis (rheumatoid)*



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Therapeutic Choices

Nonpharmacologic Choices

Treatment of red eye depends on the history, symptom presentation and working diagnosis as shown in Figure 1 and Figure 2.

Instruct patient to:

- Stop wearing contact lenses until the problem is resolved; once resolved, reassess contact lens fit and hygiene practices
- Avoid makeup, smoke, wind and other irritants
- Discard potentially contaminated eye products, including makeup
- Prevent spread through fastidious hand washing, separating towels, and avoiding school or work until infection improves (3–7 days viral) or 24 hours after starting antibiotics for bacterial conjunctivitis
- Apply cold compresses for allergic or viral conjunctivitis
- Apply warm compresses for bacterial conjunctivitis to help remove crusts, styes or chalazion to reduce swelling:
  - apply 2–3 times a day for 5–10 minutes followed by gentle massage over the lesion

- Practise lid hygiene for blepharitis:
  - place warm compresses on closed eyelids 1–2 times a day for 10 minutes (e.g., via a gel mask), followed by gentle rubbing of lid margins with warm water; a commercial eyelid wipe can also be used after the compress

Pharmacologic Choices

Once major/serious conditions are ruled out, treatment can be initiated. The choice of treatment depends on the underlying cause (see Figure 2 as well as Table 4, Table 5, Table 6, Table 7 and).

Viral conjunctivitis does not usually require treatment;[4] however, cold compresses and **ocular lubricants** may be used.

Since most cases of acute bacterial conjunctivitis are self-limiting and benign, broad-spectrum antibiotic eye drops are usually not required, but may result in a modest shortening of recovery time.[6][7] Ophthalmic corticosteroids and antibiotic/corticosteroid combinations should be avoided.[7] All **ophthalmic antibiotics** appear to have similar efficacy.[6] Whether treated or not, if no improvement is seen after 3–5 days, refer the patient to an ophthalmologist. Patients presenting with chronic, unilateral conjunctivitis should be referred to an ophthalmologist for investigation of a retained foreign body or nasolacrimal duct obstruction, which may create a reservoir for bacteria.

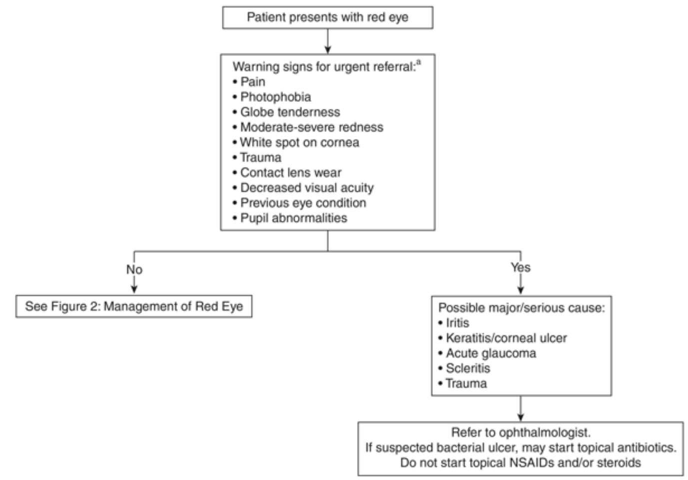
Reserve **ophthalmic vasoconstrictors** (naphazoline, tetrahydrozoline, phenylephrine) for occasional and short-term use (e.g., ≤3–4 times per month and ≤3 days in a row) to provide relief of symptoms such as redness, edema (allergic or viral conjunctivitis) or minor eye irritation. Overuse of these products may cause rebound hyperemia and/or tachyphylaxis. **Brimonidine**, a selective alpha2-adrenergic receptor agonist typically used for glaucoma, is available without a prescription in a lower concentration (0.025%) for the treatment of red eye. Based on limited preliminary evidence, brimonidine does not appear to cause significant rebound hyperemia or tachyphylaxis.[8][9]

Therapeutic Tips

- Most topically administered eye medications are capable of causing irritation or toxicity as an adverse effect.
- Some patients may react to preservatives in artificial tears and other ophthalmic products. Lubricants without preservatives are available as minims (single-dose preservative-free containers) or in multidose bottles. These formulations are preferable for patients who require drops more than 4–5 times daily for dry eye disease, those with allergic conjunctivitis or those who experience eye irritation with preserved medication.[10] Moxifloxacin is a non-preserved topical antibiotic.
- Use of ophthalmic corticosteroids may cause glaucoma and/or cataracts; the risk increases with prolonged use.
- Ophthalmic corticosteroids and antibiotic/corticosteroid combinations may worsen herpetic/fungal keratitis and should be used only under the supervision of an ophthalmologist.
- Non-steroidal anti-inflammatory drugs (NSAIDs) can cause keratitis, corneal melting and perforation in patients who have pre-existing corneal disease and, in these cases, should be used under the supervision of an ophthalmologist.[11]
- Topical decongestants/vasoconstrictors may provoke angle-closure glaucoma in patients who are predisposed (see Glaucoma).
- Topical anesthetic eye drops should never be prescribed for self-administration by the patient.

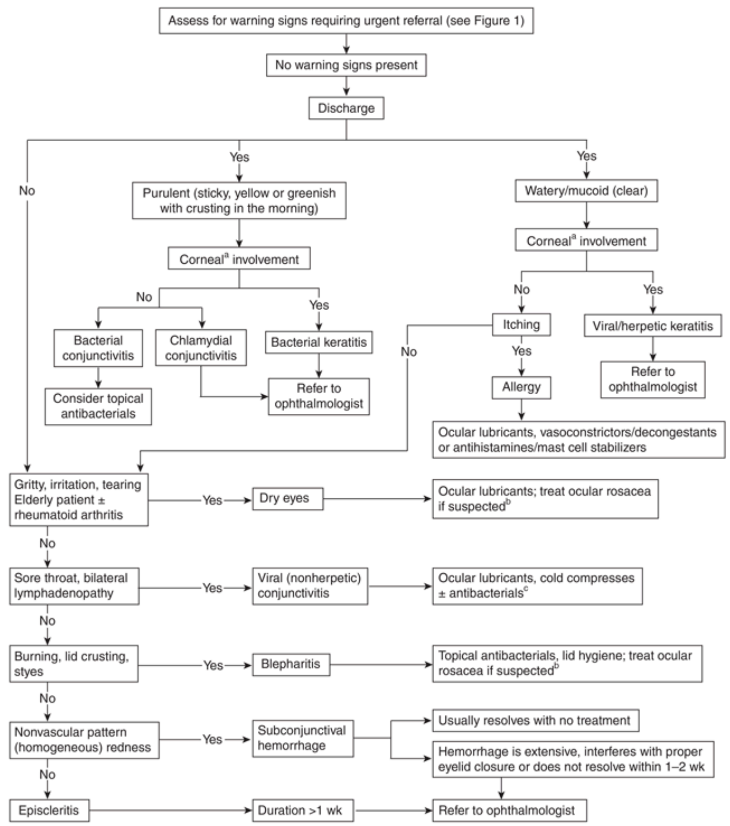
Algorithms

Figure 1: *Assessment of Red Eye*



[a] See Table 2.

Figure 2: *Management of Red Eye*



[a] Corneal staining with fluorescein strip indicates corneal involvement.
[b] See Rosacea.
[c] Used rarely to prevent secondary bacterial infection.

Drug Tables

Table 4: Anti-infectives for Red Eye

Drug/Cost[a]	Indications	Dosage	Adverse Effects	Comments
Drug Class: Antibacterials, ophthalmic				
besifloxacin Besivance $10–20	Bacterial conjunctivitis and corneal ulcers. See comments.	1 drop TID × 5–7 days	Chronic use may cause corneal epithelial toxicity, allergy and bacterial resistance.	Fluoroquinolones should be reserved for serious corneal infections.
ciprofloxacin Ciloxan, generics < $10	Bacterial conjunctivitis and corneal ulcers. See comments.	Drops: 1–2 drops Q2H while awake × 2 days then 2 drops Q4H while awake × 5 days Ointment: 1.25 cm TID × 2 days then BID × 5 days	Chronic use may cause corneal epithelial toxicity, allergy and bacterial resistance.	Fluoroquinolones should be reserved for serious corneal infections.
erythromycin generics $20–30	Superficial bacterial infections of the conjunctiva or cornea. Blepharitis and associated dry eye.	Ointment: 1.25 cm 2–6 times/day x 7–14 days	Chronic use may cause corneal epithelial toxicity, allergy and bacterial resistance.	
fusidic acid Fucithalmic $10–20	External bacterial infection of the eye. Blepharitis and associated dry eye. See comments.	1 drop Q12H × 7 days	Chronic use may cause corneal epithelial toxicity, allergy and bacterial resistance.	Due to its broad-spectrum and BID dosing, fusidic acid is especially useful in children and elderly patients. Enterobacteriaceae and pseudomonas are resistant to fusidic acid.
gatifloxacin Zymar, generics < $10	Bacterial conjunctivitis and corneal ulcers. See comments.	First 2 days: 1 drop in affected eye(s) Q2H while awake, up to 8 times daily Days 3–7: 1 drop QID	Chronic use may cause corneal epithelial toxicity, allergy and bacterial resistance.	Fluoroquinolones should be reserved for serious corneal infections.
gramicidin/polymyxin B Polysporin Pink Eye Drops, generics	Bacterial conjunctivitis/keratitis;	Drops: initially 1–3 drops Q1H, gradually	Chronic use may cause	Limit use to 4–5 days.

Drug/Cost[a]	Indications	Dosage	Adverse Effects	Comments
$10–20	blepharitis/styes. See comments.	decreasing to 1–2 drops 4–6 times daily x 7 days	corneal epithelial toxicity, allergy and bacterial resistance.	Choose broad-spectrum agent first, guided by patient allergies.
moxifloxacin Vigamox, generics $10–20	Bacterial conjunctivitis and corneal ulcers. See comments.	1 drop TID × 7 days	Chronic use may cause corneal epithelial toxicity, allergy and bacterial resistance.	The only preservative-free topical antibacterial agent Fluoroquinolones should be reserved for serious corneal infections.
ofloxacin Ocuflox $10–20	Bacterial conjunctivitis and corneal ulcers. See comments.	1–2 drops Q2–4H × 2 days then QID × 5 days	Chronic use may cause corneal epithelial toxicity, allergy and bacterial resistance.	Fluoroquinolones should be reserved for serious corneal infections.
polymyxin B/trimethoprim Polytrim, generics $30–40	Acute bacterial conjunctivitis, blepharitis, blepharoconjunctivitis.	Initially 1–3 drops Q1H, gradually decreasing to 1–2 drops 4–6 times daily	Chronic use may cause corneal epithelial toxicity, allergy and bacterial resistance.	Not currently available in Canada due to supply issues. May be available intermittently. Consult with manufacturer or wholesaler.
tobramycin Tobrex, generics < $10	External bacterial infections of the eye.	Drops: 1–2 drops Q4H Ointment: 1.25 cm BID–TID	Chronic use may cause corneal epithelial toxicity, allergy and bacterial resistance.	
Drug Class: Antibacterial/Corticosteroid Combinations, ophthalmic				
tobramycin 0.3%/dexamethasone 0.1% Tobradex $10–20	Severe blepharitis, rosacea-associated eyelid disease, post-surgical or severe conjunctivitis.	Suspension: 1 drop TID–QID Ointment: 1.25 cm ointment to conjunctival sac or lid margin TID–QID	Chronic use of ophthalmic antibacterials may cause corneal epithelial toxicity, allergy and bacterial resistance. Corticosteroids may cause minor stinging on instillation; may worsen herpetic/fungal keratitis; long-term use may cause glaucoma, cataracts.	Corticosteroids typically not recommended for the management of infectious conjunctivitis since they may worsen herpetic/fungal pathologies. May be considered on the advice of an eye-care professional for short-term use (1 wk or less).[12]
neomycin/polymyxin B/dexamethasone 0.1% Maxitrol $10–20	Severe blepharitis, rosacea-associated eyelid disease, post-surgical or severe conjunctivitis.	Suspension: 1 drop TID–QID Ointment: 1.25 cm ointment to conjunctival sac or lid margin TID–QID	Chronic use of ophthalmic antibacterials may cause corneal epithelial toxicity, allergy and bacterial resistance. Corticosteroids may cause minor stinging on instillation; may worsen herpetic/fungal keratitis; long-term use may cause glaucoma, cataracts.	Corticosteroids typically not recommended for the management of infectious conjunctivitis since they may worsen herpetic/fungal pathologies. May be considered on the advice of an eye-care professional for short-term use (1 wk or less).[12]
framycetin sulfate/gramicidin/dexamethasone Sofracort $10–20	Severe blepharitis, rosacea-associated eyelid disease, post-surgical or severe conjunctivitis.	Suspension: 1 drop TID–QID	Chronic use of ophthalmic antibacterials may cause corneal epithelial	Corticosteroids typically not recommended for the management of infectious conjunctivitis since

Drug/Cost[a]	Indications	Dosage	Adverse Effects	Comments
			toxicity, allergy and bacterial resistance. Corticosteroids may cause minor stinging on instillation; may worsen herpetic/fungal keratitis; long-term use may cause glaucoma, cataracts.	they may worsen herpetic/fungal pathologies. May be considered on the advice of an eye-care professional for short-term use (1 wk or less).[12]
Drug Class: Antivirals, systemic				
acyclovir Zovirax Oral, generics $30–50	Herpes zoster, herpes simplex.	Herpes zoster: 800 mg 5 times/day PO × 7 days Herpes simplex: 400 mg 5 times/day PO × 7–14 days	GI upset (uncommon); well tolerated.	
famciclovir Famvir, generics $20–30	Herpes zoster, herpes simplex.	Herpes zoster: 500 mg TID PO × 7 days Herpes simplex: 250 mg TID PO × 7 days	GI upset (uncommon); well tolerated.	
valacyclovir Valtrex, generics $20–30	Herpes zoster, herpes simplex.	Herpes zoster: 1 g TID PO × 7 days Herpes simplex: 500 mg TID PO × 7 days	GI upset (uncommon); well tolerated.	

[a] Cost of smallest available pack size or duration of treatment; includes drug cost only.
Dosage adjustment may be required in renal impairment; see Dosage Adjustment in Renal Impairment.
Eligible for prescribing by pharmacists as part of the Ontario minor ailments program.
Abbreviations: GI = gastrointestinal

Table 5: Ocular Lubricants for Red Eye

Drug/Cost[a]	Indications	Dosage	Adverse Effects	Comments[b]
Drug Class: Ocular Lubricants				
carboxymethylcellulose Celluvisc, Refresh Liquigel, Refresh Optive Advanced, Refresh Optive Fusion, Refresh Tears, Theratears, others < $10	Dry eyes, blepharitis, conjunctivitis, exposure, lid malpositions, minor irritations.	1–2 drops TID-QID	Preservative toxicity, gels cause filmy/blurry vision.	Preservative: sodium chlorite (Refresh products), sodium perborate (Theratears). Celluvisc does not contain preservatives. Theratears contains electrolytes to mimic composition of natural tears.
dextran 70/hypromellose (hydroxypropyl methylcellulose) Bion Tears, Tears Naturale, others < $10	Dry eyes, blepharitis, conjunctivitis, exposure, lid malpositions, minor irritations.	1–2 drops TID-QID	Preservative toxicity, gels cause filmy/blurry vision.	Preservative: polyquaternium-1 (Tears Naturale) Bion is preservative-free. Also contains electrolytes to mimic composition of natural tears.
hypromellose (hydroxypropyl methylcellulose) Genteal, Isopto Tears < $10	Dry eyes, blepharitis, conjunctivitis, exposure, lid malpositions, minor irritations.	1–2 drops TID-QID	Preservative toxicity, gels cause filmy/blurry vision.	Preservative: sodium perborate (Genteal), benzalkonium chloride (Isopto Tears).
mineral oil/petrolatum Soothe Night Time, Refresh Lacri-Lube, Systane Ointment, others < $10	Dry eyes, blepharitis, conjunctivitis, exposure, lid malpositions, minor irritations.	Ointment: 0.6 cm to inside of lower lid as needed	Preservative toxicity, gels cause filmy/blurry vision.	Ointments do not support bacterial growth and do not require preservatives. Soothe Night Time replaces previously marketed DuoLube. Systane ointment contains lanolin, which can cause ocular irritation, especially in

Drug/Cost[a]	Indications	Dosage	Adverse Effects	Comments[b]
				patients who are sensitive to wool.
polyvinyl alcohol Refresh, Tears Plus, others < $10	Dry eyes, blepharitis, conjunctivitis, exposure, lid malpositions, minor irritations.	1–2 drops TID-QID	Preservative toxicity, gels cause filmy/blurry vision.	Preservative: sodium chlorite (Refresh products).
propylene glycol-polyethylene glycol-400 Systane, Systane Balance, Systane Ultra < $10	Dry eyes, blepharitis, conjunctivitis, exposure, lid malpositions, minor irritations.	1–2 drops TID-QID	Preservative toxicity, gels cause filmy/blurry vision.	Preservative: polyquaternium-1 (Systane products).
retinol palmitate (vitamin A)/petrolatum Ocunox $10–20	Dry eyes, blepharitis, conjunctivitis, exposure, lid malpositions, minor irritations. Corneal abrasions.	Apply ointment at bedtime	Blurring of vision.	Ointments do not support bacterial growth and do not require preservatives. May contain lanolin, which can cause ocular irritation, especially in patients who are sensitive to wool. Do not use while wearing contact lenses.
sodium hyaluronate Blink, HydraSense, Hylo, I–Drop, others < $10	Dry eyes, blepharitis, conjunctivitis, exposure, lid malpositions, minor irritations.	With preservative: 1 drop up to QID; preservative-free drops can be used more frequently as needed	Preservative toxicity, gels cause filmy/blurry vision.	Preservative: sodium chlorite (Blink). HydraSense and Hylo are preservative-free.

[a] Cost of smallest available pack size; includes drug cost only.
[b] Information regarding preservatives and other non-medicinal ingredients is limited and subject to change. Always confirm ingredients on product label before recommending to patients.

Table 6: Ophthalmic Anti-allergy Agents for Red Eye

Drug/Cost[a]	Indications	Dosage	Adverse Effects	Comments
Drug Class: Antihistamines, ophthalmic				
antazoline/ naphazoline Refresh Eye Allergy Relief, others < $10	Ocular allergies.	1–2 drops Q3–4H	Minor stinging on instillation; pupillary dilation and angle-closure glaucoma in predisposed persons, e.g., elderly, female, significantly hyperopic, have a positive family history, East Asian or Inuit ancestry (potentially due to shallower anterior chamber and angle anatomy[13][14]). Ophthalmic vasoconstrictors are meant for occasional and short-term use. Overuse may cause rebound hyperemia.	Antihistamine/ophthalmic vasoconstrictor combination.
bepotastine besilate Bepreve $30–40	Ocular allergies.	1 drop BID	Minor stinging on instillation. Mild or transient taste disturbances.	Mast cell–stabilizing properties.
ketotifen fumarate Zaditor $20–30	Ocular allergies.	1 drop Q8–12H	Minor stinging on instillation.	Mast cell–stabilizing properties. Also available as a medication-releasing daily disposable contact lens.
olopatadine 0.1% Patanol, generics $20–30	Ocular allergies.	1–2 drops Q6–8H	Minor stinging on instillation.	Mast cell–stabilizing properties.
olopatadine 0.2% Pataday, generics $20–30	Ocular allergies.	1 drop daily	Minor stinging on instillation.	Mast cell–stabilizing properties.
olopatadine 0.7% Pazeo $30–40	Ocular allergies.	1 drop daily	Minor stinging on instillation.	Mast cell–stabilizing properties.
pheniramine/ naphazoline Naphcon-A, Opcon-A, Visine Advance Allergy, others	Ocular allergies.	1–2 drops up to QID	Minor stinging on instillation; pupillary dilation and angle-closure glaucoma in predisposed persons, e.g., elderly, female, significantly hyperopic, have a positive family history, East Asian or Inuit ancestry	Antihistamine/ophthalmic vasoconstrictor combination.

Drug/Cost[a]	Indications	Dosage	Adverse Effects	Comments
< $10			(potentially due to shallower anterior chamber and angle anatomy[13][14]). Ophthalmic vasoconstrictors are meant for occasional and short-term use. Overuse may cause rebound hyperemia.	
Drug Class: Mast Cell Stabilizers, ophthalmic				
sodium cromoglycate Cromolyn Eye Drops < $10	Ocular allergies.	1–2 drops 4–6 times/day	Minor stinging on instillation.	Mast cell stabilizer—does not provide immediate relief. May take 2–3 days to see symptom improvement.
lodoxamide Alomide $10–20	Ocular allergies.	1–2 drops QID	Minor stinging on instillation.	Mast cell stabilizer—does not provide immediate relief. May take 2–3 days to see symptom improvement.

[a] Cost of smallest available pack size; includes drug cost only.
Eligible for prescribing by pharmacists as part of the Ontario minor ailments program.

Table 7: Ophthalmic Anti-inflammatories for Red Eye

Drug/Cost[a]	Indications	Dosage	Adverse Effects	Comments
Drug Class: Corticosteroids, ophthalmic				
dexamethasone Maxidex < $10	Episcleritis, iritis, scleritis, some keratitis, ocular allergy.	Drops: 2 drops Q1H during the day and Q2H during the night; gradually decrease to Q3–4H then to TID-QID Ointment: small amount to conjunctival sac TID-QID	Corticosteroids: minor stinging on instillation; may worsen herpetic/fungal keratitis; long-term use may cause glaucoma, cataracts.	Corticosteroids: use only under the recommendation of an eye-care professional; monitor on a regular basis. Available as a suspension (Maxidex formulation) and solution (generic formulations).
difluprednate Durezol $20–30	Endogenous anterior uveitis.	1 drop QID in conjunctival sac of affected eye for 14 days, then taper	Corticosteroids: minor stinging on instillation; may worsen herpetic/fungal keratitis; long-term use may cause glaucoma, cataracts.	Corticosteroids: use only under the recommendation of an eye-care professional; monitor on a regular basis. More potent; reserved for more serious indications. May have more tendency to cause rise of intraocular pressure. Preservative is sorbic acid.
fluorometholone Flarex, generics < $10	Episcleritis, iritis, scleritis, some keratitis, ocular allergy.	1–2 drops BID-QID; may be used more frequently during initial 48 h if needed	Corticosteroids: minor stinging on instillation; may worsen herpetic/fungal keratitis; long-term use may cause glaucoma, cataracts.	Corticosteroids: use only under the recommendation of an eye-care professional; monitor on a regular basis. May be less likely to cause rise in intraocular pressure than other corticosteroids.
loteprednol Alrex, Lotemax $20–30	Episcleritis, iritis, scleritis, some keratitis, ocular allergy.	1 drop TID-QID for up to 14 days	Corticosteroids: minor stinging on instillation; may worsen herpetic/fungal keratitis; long-term use may cause glaucoma, cataracts.	Corticosteroids: use only under the recommendation of an eye-care professional; monitor on a regular basis. May be less likely to cause rise in intraocular pressure than other topical corticosteroids.
prednisolone Pred Forte, Minims Prednisolone, generics < $10	Episcleritis, iritis, scleritis, some keratitis, ocular allergy.	1–2 drops Q1H during the day and Q2H at night until favourable response, then 1 drop Q4H	Corticosteroids: minor stinging on instillation; may worsen herpetic/fungal keratitis; long-term use may cause glaucoma, cataracts.	Corticosteroids: use only under the recommendation of an eye-care professional; monitor on a regular basis. Minims Prednisolone is preservative-free.
Drug Class: Immunomodulators, ophthalmic				

Drug/Cost[a]	Indications	Dosage	Adverse Effects	Comments
cyclosporine (0.05%) Restasis, Restasis MultiDose, generics $45	Dry eye disease	1 drop BID	Ocular burning sensation.	Available in single-use or multidose vials (preservative-free). Allow 15 min before/after administration of ocular lubricants. Response to treatment may take up to 3 months or longer. Should be prescribed on the advice of an eye-care professional or rheumatologist.
cyclosporine (0.09%) Cequa $220	Dry eye disease	1 drop BID	Pain on instillation.	Available in single-use, preservative-free vials. Allow 15 min before/after administration of ocular lubricants. Should be prescribed on the advice of an eye-care professional or rheumatologist.
cyclosporine (0.1%) Verkazia $120	Severe vernal keratoconjunctivitis in children 4–18 y.	Initial: 1 drop QID Maintenance: 1 drop BID	Ocular pain, pruritus, foreign body sensation, headache, cough.	Available in single-use, preservative-free vials. Allow 15 min before/after administration of ocular lubricants. Should be prescribed on the advice of an eye-care professional or rheumatologist.
lifitegrast Xiidra $230[b]	Dry eye disease.	1 drop BID	Eye irritation and pain, blurred vision, dysgeusia. Blurred vision may last from minutes to a few hours.	Available only in single-use vials (preservative-free). Some patients may have initial relief after only 2 wk of treatment, although full response may take several months. Should be prescribed on the advice of an eye-care professional or rheumatologist.
Drug Class: Nonsteroidal Anti-inflammatory Drugs (NSAIDs), ophthalmic				
diclofenac Voltaren Ophtha, generics < $10	Episcleritis, minor corneal abrasions; as an adjunct to topical corticosteroids in scleritis, iritis.	1 drop 4–5 times daily	Mild to moderate stinging on instillation; epitheliopathy and possible ulceration with prolonged and frequent use.	
ketorolac Acular, Acular LS, Acuvail, generics $10–20	Episcleritis, minor corneal abrasions; as an adjunct to topical corticosteroids in scleritis, iritis.	Acular (0.5%)/Acular LS (0.4%): 1 drop QID Acuvail (0.45%): 1 drop BID	Mild to moderate stinging on instillation; epitheliopathy and possible ulceration with prolonged and frequent use.	Acuvail is available only in preservative-free single-use vials.
nepafenac Nevanac, Ilevro $20–30	Episcleritis, minor corneal abrasions; as an adjunct to topical corticosteroids in scleritis, iritis.	0.1%: 1 drop TID beginning 1 day prior to cataract surgery, continued on the day of surgery and × 2 wk postoperatively 0.3%: 1 drop once daily beginning 1 day prior to cataract surgery, continued on day of surgery and × 2 wk postoperatively	Mild to moderate stinging on instillation; epitheliopathy and possible ulceration with prolonged and frequent use. Eyelid margin crusting, eye pain, punctate keratitis, blurred vision, dry eye, pruritis, headache.	Shake well before use. Contains benzalkonium chloride as a preservative. Contact lenses should not be worn when using nepafenac.

[a] Cost of smallest available pack size; includes drug cost only.
[b] Cost may be significantly less ($60–$100) if manufacturer discount card is available.

Table 8: Ophthalmic Vasoconstrictors/Decongestants for Red Eye

Drug/Cost[a]	Indications	Dosage	Adverse Effects
Drug Class: Vasoconstrictors/Decongestants, ophthalmic			

Drug/Cost[a]	Indications	Dosage	Adverse Effects
naphazoline Clear Eyes, Refresh Redness Relief, others < $10	Alleviation of redness and/or eyelid edema in allergic or viral conjunctivitis, minor irritation (smoke, dust, wind, chlorinated pool).	1–2 drops Q3–4H PRN × 3–4 days	Minor stinging on instillation; pupillary dilation and angle-closure glaucoma in predisposed persons, e.g., elderly, female, significantly hyperopic, have a positive family history, East Asian or Inuit ancestry (potentially due to shallower anterior chamber and angle anatomy[13][14]). Ophthalmic vasoconstrictors are meant for occasional and short-term use. Overuse may cause rebound hyperemia.
brimonidine (0.025%) Lumify $10–20	Alleviation of redness and/or eyelid edema in allergic or viral conjunctivitis, minor irritation (smoke, dust, wind, chlorinated pool).	1 drop Q6–8H PRN; maximum 4x/day	Pain on instillation.
phenylephrine Mydfrin, others < $10	Alleviation of redness and/or eyelid edema in allergic or viral conjunctivitis, minor irritation (smoke, dust, wind, chlorinated pool).	1–2 drops QID PRN × ≤3 days	Minor stinging on instillation; pupillary dilation and angle-closure glaucoma in predisposed persons, e.g., elderly, female, significantly hyperopic, have a positive family history, East Asian or Inuit ancestry (potentially due to shallower anterior chamber and angle anatomy[13][14]). Ophthalmic vasoconstrictors are meant for occasional and short-term use. Overuse may cause rebound hyperemia.
tetrahydrozoline Visine, others < $10	Alleviation of redness and/or eyelid edema in allergic or viral conjunctivitis, minor irritation (smoke, dust, wind, chlorinated pool).	1–2 drops BID-QID	Minor stinging on instillation; pupillary dilation and angle-closure glaucoma in predisposed persons, e.g., elderly, female, significantly hyperopic, have a positive family history, East Asian or Inuit ancestry (potentially due to shallower anterior chamber and angle anatomy[13][14]). Ophthalmic vasoconstrictors are meant for occasional and short-term use. Overuse may cause rebound hyperemia.

[a] Cost of smallest available pack size; includes drug cost only.
Eligible for prescribing by pharmacists as part of the Ontario minor ailments program.

Suggested Readings

Smith J, Severn P, Clarke L. *BMJ best practice: assessment of red eye* [PDF file]. Available from: bestpractice.bmj.com/topics/en-gb/496/pdf/496.pdf.

American Academy of Ophthalmology. *Conjunctivitis PPP – 2018* [internet]. Available from: www.aao.org/preferred-practice-pattern/conjunctivitis-ppp-2018.

American Academy of Ophthalmology. Bacterial keratitis PPP – 2018 [internet]. Available from: www.aao.org/preferred-practice-pattern/bacterial-keratitis-ppp-2018.

Narayana S, McGee S. Bedside diagnosis of the 'red eye': a systematic review. *Am J Med* 2015;128(11):1220-4.

Pflipsen M, Massaquoi M, Wolf S. Evaluation of the painful eye. *Am Fam Physician* 2016;93(12):991-8.

References

1. Wu P, Duan F, Luo C et al. Characteristics of ocular findings of patients with coronavirus disease 2019 (COVID-19) in Hubei Province, China. *JAMA Ophthalmol* 2020;138(5):575-8.
2. Azari AA, Barney NP. Conjunctivitis: a systematic review of diagnosis and treatment. *JAMA* 2013;310(16):1721-9.
3. Kam KY, Ong HS, Bunce C et al. Sensitivity and specificity of the AdenoPlus point-of-care system in detecting adenovirus in conjunctivitis patients at an ophthalmic emergency department: a diagnostic accuracy study. *Br J Ophthalmol* 2015;99(9):1186-9.
4. Epling J. Bacterial conjunctivitis. *BMJ Clin Evid* 2012;2012:0704.
5. Wong MM, Anninger W. The pediatric red eye. *Pediatr Clin North Am* 2014;61(3):591-606.
6. Sheikh A, Hurwitz B, van Schayck CP et al. Antibiotics versus placebo for acute bacterial conjunctivitis. *Cochrane Database Syst Rev* 2012;(9):CD001211.
7. Shekhawat NS, Shtein RM, Blachley TS et al. Antibiotic prescription fills for acute conjunctivitis among enrollees in a large United States managed care network. *Ophthalmology* 2017;124(8):1099-107.
8. Ackerman SL, Tokildsen GL, McLaurin E et al. Low-dose brimonidine for relief of ocular redness: integrated analysis of four clinical trials. *Clin Exp Optom* 2019;102(2):131-9.
9. Hosten LO, Snyder C. Over-the-counter ocular decongestants in the United States - mechanisms of action and clinical utility for management of ocular redness. *Clin Optom (Auckl)* 2020;12:95-105.
10. Jones L, Downie LE, Korb D et al. TFOS DEWS II management and therapy report. *Ocul Surf* 2017;15(3):575-628.
11. Kim SJ, Flach AJ, Jampol LM. Nonsteroidal anti-inflammatory drugs in ophthalmology. *Surv Ophthalmol* 2010;55(2):108-33.
12. Holland EJ, Fingeret M, Mah FS. Use of topical steroids in conjunctivitis: a review of the evidence. *Cornea* 2019;38(8):1062-7.
13. Yip JLY, Foster PJ. Ethnic differences in primary angle-closure glaucoma. *Curr Opin Ophthalmol* 2006;17(2):175-80.
14. Wojciechowski R, Congdon N, Anninger W et al. Age, gender, biometry, refractive error, and the anterior chamber angle among Alaskan Eskimos. *Ophthalmology* 2003;110(2):365-75.